

LLM-Guided Synthesis of Constraint-Adaptive Heuristics for Satellite–Ground Scheduling

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Abstract

Graph optimization heuristics can achieve strong aggregate performance while relying on decision rules that fail under changes in resource constraints. Effective adaptation requires identifying not only when action preferences should change, but also whether the input representation contains the information needed for that change. We propose an LLM-guided synthesis framework for constraint-adaptive satellite–ground scheduling that uses certified decision evidence to jointly refine graph representations and ranking rules. Its central mechanism detects ranking requirements that become contradictory under the current feature representation and directs synthesis toward the missing structural distinctions, rather than repeatedly adjusting an insufficient scoring interface. Two complementary mechanisms support this refinement. First, controlled offline interventions on resource-constraint configurations generate paired static scheduling problems in which the same competing actions remain individually feasible. Sound bounds on conditional completion values certify preference reversals or preservation under explicit boundary commitments, yielding relational specifications of when rankings should change or remain stable. Second, structural witnesses guide the LLM to compose distinguishing features from a typed graph-operation library and synthesize corresponding ranking rules. Feature–rule pairs are selected jointly for finite input representability, preference fit, complete-schedule quality, and feature-computation cost, making representation expansion targeted and cost-aware. We formulate contact selection as maximum-weight independent set optimization and study adaptation using a frozen program on held-out scheduling instances and constraint configurations. The synthesized program responds to the current conflict graph within a fixed feasibility-enforcing kernel, without online LLM or full-residual certification-oracle calls. By linking representation refinement to certified decision conflicts, the framework addresses information limitations beyond rule-only repair while explicitly accounting for the computational cost of constraint adaptation.

CCS Concepts

• **Computing methodologies** → **Planning and scheduling**; • **Mathematics of computing** → *Combinatorial optimization*.

Keywords

Satellite scheduling, maximum-weight independent set, heuristic synthesis, representation refinement, large language models



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1 Introduction

A satellite–ground contact excludes opportunities sharing its resources. Changing a switching gap changes which alternatives can coexist, and hence their opportunity costs, while preserving contacts and rewards. We isolate the static pairwise-conflict subproblem as maximum-weight independent set (MWIS) [3, 19] and study a reusable heuristic across resource configurations.

LLM synthesis searches executable heuristics through evaluation and reflection [21, 26, 33]. Aggregate reward does not identify whether failure needs a different formula or missing input information. No pointwise rule can satisfy incompatible strict preferences over identical observations.

Figure 1 illustrates this obstruction. Two competing contacts have identical nine-input vectors within each configuration. Their forced-completion increments change from 20, 26 to 20, 14 after a ground-gap intervention, reversing preference while both contacts remain individually feasible. A base-only scalar necessarily ties them; neighborhood compatibility supplies the missing distinction. These illustrated values exclude the common fixed reward and are conditional optima, not heuristic rollout returns.

We propose *certified representation–rule refinement*. Offline interventions preserve contacts and explicit boundary commitments. Sound bounds certify full-residual preference reversals or preservation; unresolved comparisons abstain. Identical residual components cancel within each configuration. The resulting strict requirements are projected onto exact feature-equivalence classes. A directed cycle proves that every scalar on the interface is insufficient. Its concrete ranking arcs and equality joins identify occurrences that must be distinguished, adapting abstraction-refinement principles [9, 11] to optimization decisions.

We use an LLM as a conditionable proposal operator in the combinatorial space of bounded typed graph computations and score expressions. Witnesses identify necessary distinctions without supplying their computation, conditioning generation on an information deficit. Deterministic search remains viable: our supplied 53-expression TRAIN catalogue admits exact minimum-additive-cost repair. The theory motivates refinement; proposer efficiency remains empirical.

Splitting one witness can expose another, requiring complete-quotient rechecks. TRAIN selection checks acyclicity, scalar fit, scheduling quality and work before freezing. Priorities control greedy additions and pivots in shared classical repair [15, 18]; repeated deployment needs no LLM or full-residual certification calls.

Our contributions are:

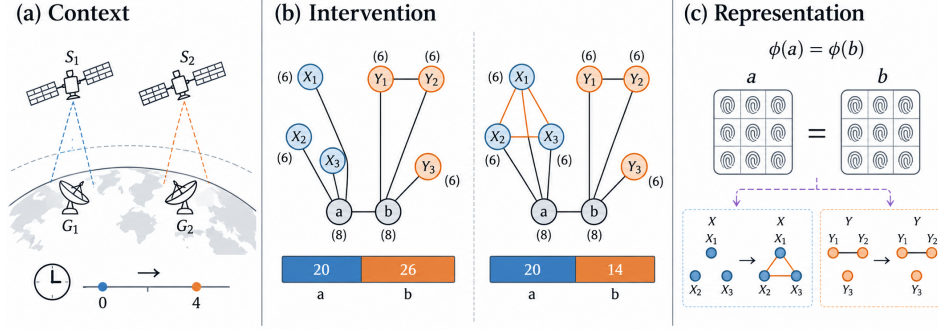


Figure 1: Constraint adaptation can require new information. Forced-completion increments of a, b change from 20, 26 to 20, 14 in this constructed example, reversing preference despite equal nine-input vectors within each configuration. The common fixed reward is excluded; structural compatibility supplies the missing distinction.

- (1) **Information diagnosis:** finite pointwise representability with concrete cycle witnesses, separating insufficient inputs from poor rule fit.
- (2) **Targeted cost-aware synthesis:** certified completion requirements and equality joins direct typed feature–rule refinement, with full-quotient rechecks and a finite-catalogue additive-cost guarantee.
- (3) **Frozen evaluation:** evidence-controlled authoring, separate representability and scalar-fit measurements under a common kernel, and separate published-solver comparisons.

Shared-seed TRAIN repair yields 26/32 information-compatible W proposals versus 19/32 with relations alone. Joint selection raises withheld fit within W by 3.37 points, while relations-only outputs retain stronger fit. Scheduling improvement remains separate and is tested against published solvers.

2 Related Work

Satellite scheduling couples observation, downlink and shared ground resources [3, 19, 25]. Graph learning guides constructive optimization and exact search [10, 12, 20]; local topology can aid neural generalization [4]. Our question concerns whether an interface can express certified decision requirements before fitting a scorer.

LLM design. FunSearch combines generated programmes with executable evaluation; EoH evolves descriptions and code; ReEvo and LLaMEA add reflective or evolutionary fitness feedback [21, 26, 29, 33]. A₂DEPT evolves complete solvers with hierarchical edits, while PathWise carries derivation history through a search-memory graph [7, 16]. Their open-ended programmes can introduce structural computations, so feature invention alone is not our contribution. We supply certified contradictions in the current interface and concrete equality joins to direct that invention. DRAGON uses decomposition and reconstruction agents during optimization [8]; here an offline LLM yields a frozen priority inside classical search. Dynamic heuristic selection and algorithm configuration optimize state-dependent choices and settings [17, 28]; we refine the available input distinctions and account for their computation.

Representation and synthesis. GNN expressiveness connects architectural distinctions to graph isomorphism tests [23, 32]. Our quotient instead collapses exactly equal observed feature vectors. Generalized planning learns features, representations and policies shared across instances [5, 6, 11]. CEGAR and abstraction-guided synthesis refine abstractions after counterexamples [9, 30]; sketching, syntax-guided and relational synthesis separate syntax from semantic obligations [1, 27, 31]. Our requirements are sound conditional preferences made contradictory by information collapse, rather than spurious abstract executions. Equality joins identify distinctions needed before rule fitting. Finite representability, bounded-grammar fit and deployment effectiveness remain different properties.

Classical MWIS and evaluation. Independent-set exchanges and clique-based branch-and-bound provide foundations [2, 24]. Published weighted solvers include concurrent CHILS search, multilevel M²WIS, Struction transformations and WeightedBR [13–15, 18]. These mature components support our common kernel and separate algorithm comparisons; they are not claimed as new search primitives. Artificial weights can favor particular heuristics and inference rules [22]. We therefore include application-derived public weighted graphs, preserving objectives through exact transformations and reporting numerical compatibility.

3 Problem Setting

A contact v specifies satellite σ_v , station γ_v , interval $[s_v, e_v]$, with $s_v < e_v$, and nonnegative reward w_v . Selection reserves that entire interval. Each main-model satellite and station has capacity one. For contacts ordered by start, a shared station conflicts if $s_v < e_u + \tau_g$; a shared satellite uses $s_v < e_u + \tau_s$. Nonnegative switching gaps are fixed within configuration θ . Synthetic reward is independent of duration. The weighted conflict graph $G_\theta = (V, E_\theta, w)$ yields

$$\max_{I \subseteq V} w(I), \quad w(I) = \sum_{v \in I} w_v. \quad (1)$$

$I \text{ independent in } G_\theta$

This models constraints expressible by pairwise conflicts. Public graphs exercise MWIS without a satellite interpretation; exploratory C3 tracks retain their separate inherited predicates.

An admissible boundary $B = (F, X)$ records independent permanent commitments F , disjoint from permanent exclusions X . Available contacts are

$$R_\theta(B) = V \setminus (F \cup X \cup N_\theta(F)). \quad (2)$$

For $d \in R_\theta(B)$, its full conditional-completion value is

$$V_\theta(d \mid B) = \max_{\substack{I \subseteq V \text{ independent in } G_\theta \\ F \cup \{d\} \subseteq I, I \cap X = \emptyset}} w(I). \quad (3)$$

It includes fixed reward and every compatible global completion. An aligned intervention changes one constraint parameter while preserving contacts, resources, intervals, weights and boundary. The common boundary must be admissible under both configurations. Competing a, b must remain available and adjacent in both configurations. Their certified preferences can change even though each remains individually feasible. Adaptation compares separate static problems, not online constraint changes.

A frozen deterministic typed programme $P = (\phi_P, h_P)$ gives finite priorities $h_P(\phi_P(G_\theta, A, v))$ on active snapshot A . All programmes share an initializer, patch rules, classical search and feasibility checks. They alter greedy additions and pivots, which can change later repair trajectories. The kernel protects F , avoids X , retracts only movable incumbent vertices and commits only globally feasible positive gains. Budget termination retains a feasible incumbent, not necessarily an optimum. Full-residual labels and restricted-patch search have different temporary commitments; ranking agreement and schedule quality require separate evaluation.

4 Certified Representation–Rule Refinement

Figure 3 separates certified preferences, quotient-guided feature–rule synthesis and frozen execution in a common bounded kernel.

4.1 Full-Residual Decision Evidence

At boundary $B = (F, X)$, put $A = R_\theta(B)$ and $A_d = A \setminus (\{d\} \cup N_\theta(d))$. Let C_d be the connected components of $G_\theta[A_d]$, identified by complete vertex sets. Then

$$V_\theta(d \mid B) = w(F) + w_d + \sum_{C \in C_d} \alpha_w(G_\theta[C]), \quad (4)$$

where α_w is the MWIS optimum. Within one graph and boundary, set $\mathcal{A} = C_a \setminus C_b$ and $\mathcal{B} = C_b \setminus C_a$; identical components cancel:

$$\Delta_{ab} = w_a - w_b + \sum_{C \in \mathcal{A}} \alpha_w(G_\theta[C]) - \sum_{C \in \mathcal{B}} \alpha_w(G_\theta[C]). \quad (5)$$

Sound component intervals $[L_C, U_C]$ give

$$\begin{aligned} \underline{\Delta}_{ab} &= w_a - w_b + \sum_{C \in \mathcal{A}} L_C - \sum_{C \in \mathcal{B}} U_C, \\ \bar{\Delta}_{ab} &= w_a - w_b + \sum_{C \in \mathcal{A}} U_C - \sum_{C \in \mathcal{B}} L_C. \end{aligned} \quad (6)$$

Lemma 1 (within-graph cancellation). The endpoints enclose the full conditional-completion difference. *Proof.* Component optima unite feasibly; identical induced components cancel. Bounds on remaining signed terms give the enclosure. $\square$

Cancellation requires identical vertex sets within the same graph, not equal size or weight; overlapping components remain separate.

Verified packings give lower bounds and clique partitions give upper bounds by summing clique maximum weights. Bounded exact search tightens intervals; unsearched components retain $[0, w(C)]$. Rational arithmetic and outward display rounding preserve soundness; unresolved comparisons abstain.

Certify aligned configurations separately. For $\epsilon \geq 0$, $\frac{\theta_0}{\Delta_{ab}} > \epsilon$ and $\frac{-\theta_1}{\Delta_{ab}} < -\epsilon$ certify reversal; equal strict signs certify preservation. Ties impose no strict requirement. Certified pairs require

$$h(\phi(x^+)) > h(\phi(x^-)), \quad x = (G_\theta, A, v). \quad (7)$$

Each label retains its graph, boundary, actions and signed bounds.

4.2 Complete Quotients and Structural Witnesses

Merge occurrences with exactly equal represented values under frozen semantics. Quotient Q_ϕ has one vertex per class and an arc $[x^+] \rightarrow [x^-]$ for every strict requirement, including self-loops. The base nine coordinates are reward, duration, degree, neighbor-weight sum/maximum, compatible-weight sum, ground/satellite gaps and residual count; explicit IDs and dataset labels are excluded.

Theorem 1 (finite pointwise representability). An unrestricted deterministic scalar $h(\phi(x))$ satisfies all observed requirements if and only if Q_ϕ is acyclic. *Proof.* A directed cycle demands a strict decrease back to the same vector. On a finite DAG, each class’s longest outgoing-path length strictly decreases along every arc. Scaling gives any positive margin. $\square$

Observed-vector existence implies neither bounded-grammar expressibility nor unseen correctness. A cycle stores concrete arcs $p_i \rightarrow n_i$ and equality joins $n_i \sim_\phi p_{i+1}$, with cyclic indices (Figure 2); a repair must split a join. Joins can cross configurations or boundaries, making statewise checks insufficient. Splitting one witness can expose another; each proposal rebuilds the *complete* quotient over all retained requirements.

4.3 Scoped Minimum-Cost Interface Repair

Fix the base, evidence and occurrence feature table. For finite catalogue $\mathcal{F}$, positive standalone costs c_f and limit K , set $d_{Wf} = 1$ when f splits a join of witness W . The optional exact master is

$$\begin{aligned} \min_{z \in \{0,1\}^{|\mathcal{F}|}} \quad & \sum_f c_f z_f, \\ \text{subject to} \quad & \sum_f d_{Wf} z_f \geq 1 \quad (\text{retained } W), \\ & \sum_f z_f \leq K. \end{aligned} \quad (8)$$

Globally solve each master and rebuild the entire quotient from the base and selected additions. A surviving cycle adds a cut; subsequent solutions can replace additions. Refined cycles can revisit coarse classes, so splitting only the original witness is insufficient.

Theorem 2 (finite-catalogue additive optimum). An acyclic terminating solution minimizes additive cost among catalogue repairs satisfying K . *Proof.* Every acyclic repair splits a join of each retained witness, satisfying all cuts. Exact master cost lower-bounds valid repairs; the final DAG attains it. Each cut excludes the current cyclic

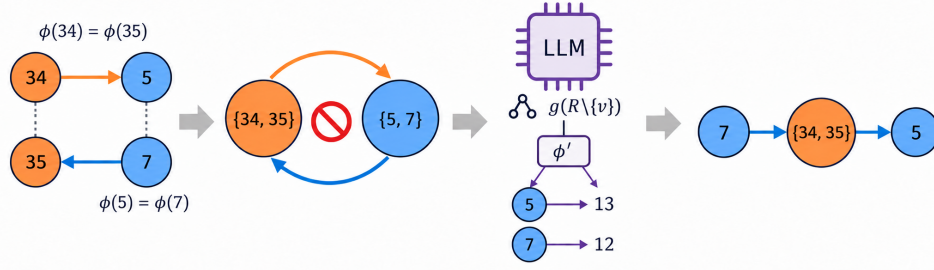


Figure 2: A structural witness tells synthesis what to distinguish. In a real TRAIN seed, requirements $34 > 5$, $7 > 35$ and joins $5 \sim 7$, $34 \sim 35$ create a two-cycle. Replacing a coordinate with the recorded cold-bank root-deletion feature separates 5, 7 (feature values 13, 12), removing this obstruction. Joint selection still rechecks the complete inventory.

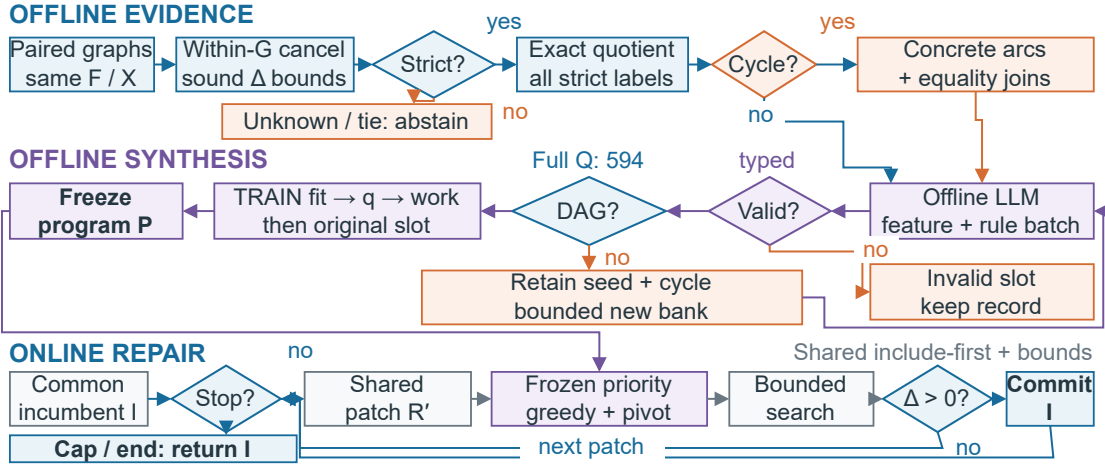


Figure 3: Offline refinement and shared deployment. Quotient witnesses guide typed proposals; joint selection checks representability, scalar fit, reward and work before freezing. Priorities guide greedy additions and pivots within common feasible classical search. R2 tests one bounded warm-repair round of this general procedure.

subset. Finitely many subsets imply termination or infeasibility under unrestricted search. $\square$

Inseparable witnesses and truncated masters remain unresolved. This guarantee covers fixed evidence and additive standalone work, not shared-DAG runtime or scalar fit. Practical proposal banks do not inherit master optimality.

4.4 Typed Feature–Rule Proposals

Why use an LLM? A witness diagnoses joins to split without constructing a distinguishing computation and compatible score. Typed graph operations, reductions and score expressions create a combinatorial programme space. The LLM supplies a witness-conditioned proposal operator [21, 26]; deterministic quotient, scalar-fit and execution checks retain authority. Theorem 1 requires representation refinement, not an LLM. The fixed-catalogue procedure is a deterministic alternative once expressions are available.

The offline grammar permits active, neighbor, root-deleted and compatible continuation sets, typed union/intersection/difference,

and count, weight-sum, edge-count, greedy-packing and clique-cover reductions. Arithmetic and conditionals combine scalars. Static validation enforces types, bounded syntax and deterministic operations rather than unrestricted executable code.

Constructing a distinguishing programme. The request supplies graph snapshots, certified relations, concrete joins, the current programme and typed grammar. For an illustrative construction, not a learned winner, let $D_v = A \setminus \{v\}$ and $C_v = A \setminus (N(v) \cup \{v\})$. Valid packing and induced-edge reductions give

$$\begin{aligned} f_p(v) &= g(C_v), & f_e(v) &= |E[D_v]|, \\ h(v) &= w_v + \alpha f_p(v) - \beta f_e(v) / (1 + \deg(v)), \end{aligned} \quad (9)$$

where g is greedy-packing weight and α, β are bounded constants. Continuation packing measures coexistence; root deletion exposes residual structure. Actual join splitting and score signs still require complete-inventory checks. The LLM constructs expressions and rules together; the catalogue master selects existing expressions.

The information gate uses all nine base coordinates and only added features referenced by the rule; unused additions cannot repair it. All declarations must pass finite-value checks. Shared expression DAGs and branch-sensitive evaluation avoid redundant work. Priorities are computed once per immutable patch snapshot; initialization and feature queries are charged.

4.5 Shared Classical Execution and Joint Selection

All conditions share Degree initialization or a charged CHILS incumbent [15], then a bounded clique-bound repair kernel. Feasible incumbents, weighted bounds and local improving replacements are standard MWIS primitives [13, 14, 18]. These shared components are not our contribution. Imported incumbents must be independent, include permanent F and exclude X . For independent incumbent I , destroy movable $D \subseteq I \setminus F$ and retain $F_{\text{out}} = I \setminus D$. A patch satisfies $D \subseteq R' \subseteq V \setminus (F_{\text{out}} \cup N(F_{\text{out}}) \cup X)$. Branch-and-bound searches independent replacements in R' ; only checked positive exact gains are committed. The programme controls greedy additions and pivots; targets, patches, bounds and include-first traversal remain common. Caps retain the feasible incumbent. Full-residual labels do not certify restricted-patch preferences.

Eligibility requires finite validated syntax, an acyclic demanded quotient and feasible error-free TRAIN execution. Strict fit $m(P)$ counts requirements whose score difference exceeds $\eta \geq 0$. For family state sets $\mathcal{D}_t$, use

$$q(P) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \frac{1}{|\mathcal{D}_t|} \sum_{G \in \mathcal{D}_t} \frac{w(I_P(G))}{w(V_G)}. \quad (10)$$

This is total-input-reward fraction, not optimum percentage; zero-total states use one. Feature-plus-repair work $c(P)$ uses the same averaging. Eligible selection maximizes m , then q , minimizes c , then uses original slot. Empty banks have no replacement. Quality-only baselines maximize q , then minimize c , without a gate. Budgets, ASTs and selectors freeze on TRAIN. Measured CPU/wall accompany the work proxy; deployment calls neither LLM nor full-residual certification oracle.

5 Experiments

We distinguish representability, scalar fit and scheduling performance. All conditions share the typed library and classical kernel; shared-search gains alone do not establish LLM-guidance benefit.

5.1 Decision Evidence and Authoring Design

TRAIN contains 120 unit-weight states with at most 32 vertices: 24 source-induced public graphs and 96 contact endpoints forming 48 intervention pairs. Public clusters are partitioned within a previously exposed corpus. Contact pairs preserve opportunities and change only ground switching gap, $0 \rightarrow 1$; satellite gap remains zero and capacities one. Queries precede certification and retain shortfalls. Of 1,503 comparisons, 594 are strict, 909 tie and none are unknown; 90 strict comparisons alias under base-nine vectors. Among 630 paired queries, 12 reverse and 123 preserve strict preferences, 294 tie on both sides and 201 have one tied endpoint. This controlled, alias-enriched frame diagnoses conflicts, not their natural incidence.

Table 1: R2 information-gate yield: eligible/original positions in all five blocks and the preregistered matched four. Bold marks observed column bests.

Arm	All five	Matched four
W	28/40	26/32
R	24/40	19/32
O	0/40	0/32

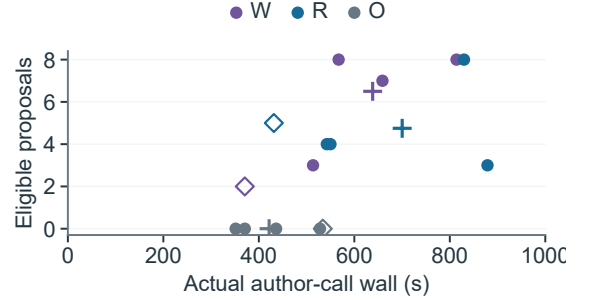


Figure 4: LLM proposal efficiency: eligible proposals versus actual author-call wall time. All five banks remain shown; the fifth is hollow and matched means are crosses. Times describe whole calls.

R2 repairs a shared rejected interface and cycle. Prompts show six cases and eleven relations; checking retains all 594 requirements. Five blocks contain eight proposals per W/R/O arm; the first four form the preregistered matched cohort and the fifth remains reported. Authoring requests gpt-6.1-sol/ultra; the served backend is undisclosed. All 120 programs complete server TRAIN assessment with feasible incumbents. Matched slots do not match compute.

5.2 What Witness Feedback Repairs

Matched information-gate yield is 26/32 W versus 19/32 R, a 21.875-point difference (Table 1). The table also retains the complete five-block cohort; Figure 4 plots all five bank counts per arm against measured author-call wall cost. These are conditional interface repair findings, not general model efficacy.

Matched W/R banks produce 10.18/6.78 eligible proposals per 1,000 measured author-call seconds, descriptively conditional on this seed and these calls. The gate measures whether the demanded features admit a consistent ordering; it does not certify the authored scalar or scheduling benefit. Detailed scalar fit, feature work and policy quality remain in the supporting report and Discussion, separate from information repair. All TRAIN schedule qualities equal 2693/4480 under Eq. 10; there is no observed TRAIN reward improvement, before any distribution shift.

Deployment freezes four genuine W joint and twelve matched quality-only selections before TEST certification; evaluation outcomes select no winner. Joint-versus-quality contrasts include selection, not isolated witness effects.

5.3 Complete Quotient Checking and Feature Cost

The optional catalogue contains all 53 canonical expressions from valid cold authoring, adaptively discovered on TRAIN, with the same 594 requirements and $K = 6$. Its 4/8/16-expression prefixes remain cyclic. The full catalogue resolves the quotient using four features at exact minimum additive cost 1,371,468; a verified non-negative weighting of necessary cuts proves the matching lower bound.

Seven self-loop cuts precede two cycle cuts: removing self-loops leaves a cycle, and splitting it exposes another, requiring the ninth full-inventory check (Figure 5). Greedy tie orders remain identifier-sensitive. This certifies scoped representation repair and an additive feature-work proxy, not a fitted scalar or runtime optimum.

5.4 Published Solvers and Common Components

Stochastic native solvers use three fixed seeds; deterministic methods and Degree use one. Exact transformations preserve objectives; signed-range incompatibilities remain nulls. Coverage and actual costs accompany rewards; nominal targets are not end-to-end hard deadlines.

EoH-DSL [21] adapts published operators in four runs of eight genuine calls with a two-member distinct-quality population. The typed interface and common rejected seed delimit this reproduction. Its four frozen deployment identities retain the shared seed; full TRAIN fitness and authoring cost are reported in the supporting material and Discussion. Authoring compute is not matched to R2.

5.5 Adaptation to Withheld Constraint Configurations

The mechanism TEST contains 24 source-disjoint public states and 48 newly drawn contact endpoints forming 24 pairs. Contact generation retains TRAIN’s resource regimes and unit weights, but uses new seeds and withheld gaps $0.25 \rightarrow 2$. Within a pair, the same contacts induce two conflict graphs. All four W-joint rules omit the available gap coordinates and recompute structural features from the current graph. No TEST preference label enters their scoring. Certification occurs after the program freeze and supplies evaluation targets only.

We evaluate 31 frozen identities and five prescribed identifier renamings. Of 962 comparisons, 453 are strict (including 79 base-nine aliases) and 509 are exact ties. The 355 paired queries comprise 13 strict reversals, 83 strict preservations, 130 ties on both endpoints and 129 with one tied endpoint. All 190 quota shortfalls remain recorded. Programs and selections freeze before evaluation certification; no TEST outcome selects a winner.

Across four blocks, W joint attains mean original strict/base-alias fit of 91.94/81.96%. Its strict advantage over W quality is 3.37 percentage points (source-cluster 95% interval: 2.04–4.72). Joint-versus-quality comparisons include selection; intervals condition on the frozen programs, not model-population effects. All four original demanded feature quotients are acyclic. This evaluates scoped information repair and actual scalar fit separately.

Table 2: Ranking adaptation to gap $0.25 \rightarrow 2$ with identical contacts: all 13 reversal and 83 preservation requirements, original mapping. LLM rows average four dependent frozen outputs; Degree has one. EoH [21] retains a shared seed AST. Counts are output–requirement positions, not independent sources or scheduling gains.

Frozen cohort	Reversal (%)	Preservation (%)
W joint	82.69 (43/52)	95.48 (317/332)
W quality	82.69 (43/52)	89.76 (298/332)
R quality	86.54 (45/52)	97.89 (325/332)
O quality	69.23 (36/52)	92.17 (306/332)
EoH-DSL	84.62 (44/52)	100.00 (332/332)
Degree	46.15 (6/13)	85.54 (71/83)

Both endpoints must satisfy their certified strict requirements for paired correctness. Mean W joint reversal/preservation correctness is 82.69/95.48% (Table 2). These targets concern configuration response, rather than arbitrary public-graph rankings. Figure 6 retains every predeclared main identity and its complete five-rename mean; no outcome selects plotted programs. Renaming is a separate stress test, not an invariance guarantee. The four EoH identities retain their seed origin and remain dependent.

All 13,392 shared-kernel assignments complete with feasible incumbents. The supporting report retains all 31 identities, full cohort contrasts, renaming obstructions, rewards and measured costs; Discussion examines their limits. These transfer measures characterize information repair; scheduling quality is assessed separately.

5.6 From Full-Residual Evidence to Restricted Patches

A TRAIN bridge examines first patches of 120 common-Degree traces; 106 contain only two vertices. Degree fits all 15 local strict labels; W joint fits 58/60 output–label positions. Of seven full-strict pairs, one reverses and three tie under the outside-fixed patch boundary. Thus full-residual fit cannot certify patch preference or search efficiency. Complete rows, comparator fits and shortfalls remain in the report.

5.7 Published-Solver Comparison and Joint-Shift Stress

We evaluate all 23 frozen policy identities on 302 contexts at nominal wall targets $T \in \{0.1, 1, 5\}$ seconds. Newly drawn scheduling problems provide 216 endpoints from 108 resource-intervention pairs. Public inputs comprise 25 WDP, three Segmentation, and ten Grids sources from the weighted-clique benchmark [22], retaining exact weights and complement orientation. Public source clusters follow fixed withholding within previously exposed corpora. Two 24-context C3 tracks remain separate exploratory analyses.

Table 3 displays four predeclared families at $T = 5$ against CHILS and ILS [15], M2WIS [14], Struction [13], and WeightedBR [18]. The deployment comparison keeps the same strong CHILS incumbent and repair limits for joint witness, Degree, and EoH-DSL policies. Four frozen EoH-DSL outputs retain their common seed AST [21]. Every warm policy is charged the full shared initializer and its own

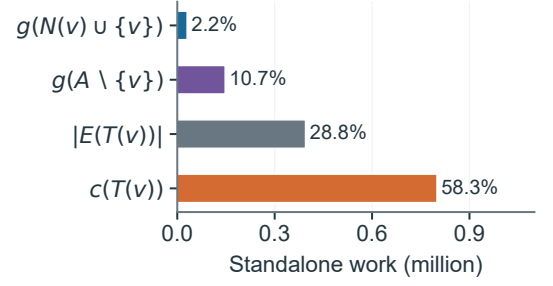
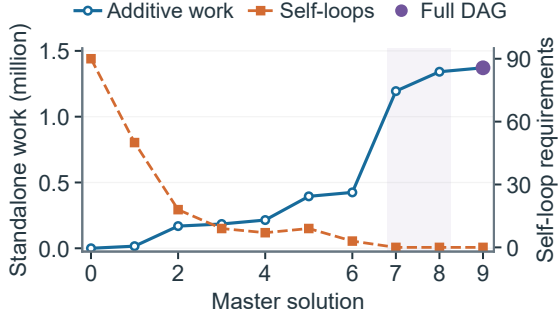


Figure 5: TRAIN catalogue repair: full-quotient obstructions and additive cost (left), minimum-cost feature work (right). Cyclic trials are not repairs. Standalone work excludes shared-DAG savings and complete-policy cost. $g(S)$ is greedy-packing weight, $T(v) = A \setminus (\{v\} \cup N(v))$, and $c(S)$ is clique-cover upper-bound weight.

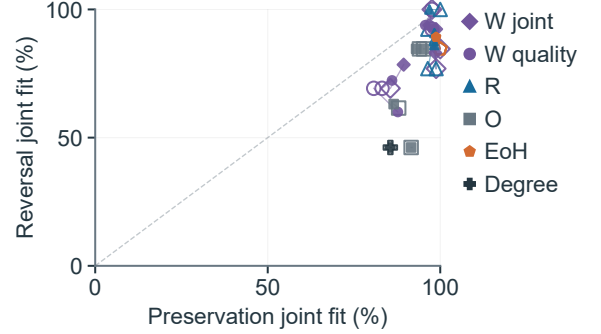
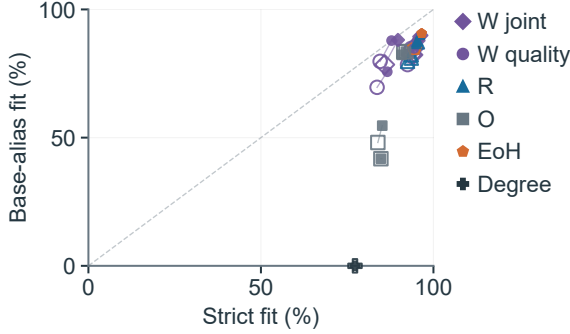


Figure 6: Frozen preference transfer: strict versus base-alias fit (left), strict-preservation versus strict-reversal correctness (right). Hollow/solid points denote original/five-rename means of all 21 predeclared identities; coincident identities and renamings remain dependent.

Table 3: Published solvers and frozen deployment at nominal $T = 5$: within-family mean objective and measured standalone CPU seconds. WDP objectives are displayed in millions. Superscripts mark defined/assigned contexts; a partial reward mean cannot establish superiority over a fully covered method. Unsupported native executions have no displayed cost; other costs include failed attempts, charged initialization and repair, with graph loading separate. Bold marks lowest observed mean CPU among fully covered methods. All 13 families and targets remain in the report.

Method	Standard/balanced		Dense/ground		WDP 4xx		Grids	
	$w(I)$	CPU	$w(I)$	CPU	$w(I)/10^6$	CPU	$w(I)$	CPU
CHILS [15]	7,277.9	5.01	978.2	5.08	75.5735	5.17	59,093.5	5.05
ILS [15]	7,277.9	5.02	978.2	5.09	75.5735	5.23	59,039.6	5.06
M ² WIS [14]	7,277.9	0.19	978.2	1.01	— ^{0/6}	—	— ^{0/10}	—
Struction [13]	7,277.9	0.03	978.2	0.21	— ^{0/6}	—	— ^{0/10}	—
WeightedBR [18]	7,277.9	0.02	977.5	3.59	— ^{0/6}	—	— ^{0/10}	—
Degree / CHILS	7,277.9	3.30	978.2	3.56	75.5735	2.81	59,077.4	4.75
EoH / CHILS [21]	7,277.9	3.88	978.2	4.88	75.5735	3.12	59,077.4	4.96
Joint W / CHILS	7,277.9	4.25	978.2	5.03	75.7194 ^{5/6}	3.29	— ^{0/10}	4.45

repair cost; nominal targets are not equal end-to-end deadlines across C++ and compiled Python.

Fresh contact pairs change ground gap $0.5 \rightarrow 6$ with unchanged opportunities. They also increase size to 512/1,024/2,048 contacts

and vary weights, durations and density. This combined shift tests deployment stress; it cannot isolate the effect of unseen gaps. Arbitrary public graphs test solver compatibility, not satellite constraint adaptation.

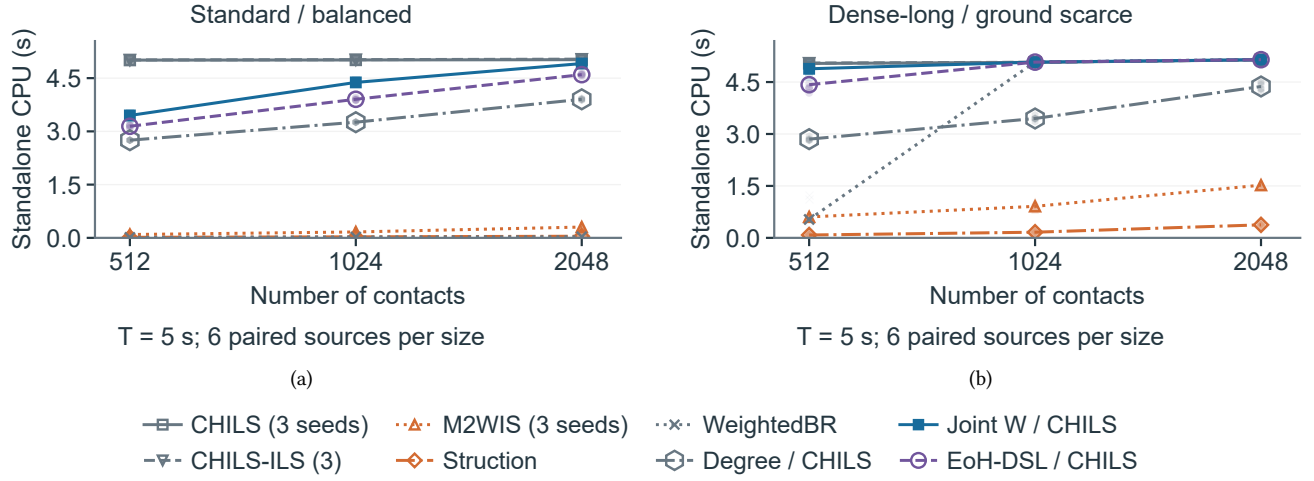


Figure 7: Measured deployment CPU versus contact count at fixed nominal $T = 5$ seconds: standard balanced (a) and dense-long ground scarce (b). Faint points show six paired sources per size; lines show their mean, after within-context seed/member averaging. Warm policies each charge the full common CHILS initializer and their own repair; shared graph loading remains separate. These are scale comparisons, not anytime traces or evidence of LLM speed superiority. Complete gain/coverage matrices remain in the supporting report.

Figure 7 plots measured CPU against size at fixed $T = 5$ for two predeclared contact families. Points retain paired-source costs; lines join their means. Complete cold/warm reward, coverage and all 13-family/three-target matrices remain in the supporting report. Cohort reward means require every member to return a successful feasible schedule; partial means cannot establish superiority over fully covered comparators. No test outcome changes a program, budget or displayed family.

The common optimization backbone retains its feasible incumbent while the synthesized policies provide refined decision representations. The deployment results illustrate this separation between information adequacy and achieved schedule quality; they do not establish an incremental LLM quality gain.

6 Discussion

Representation adequacy, scalar fitting and search effectiveness are distinct. Repairing a quotient enlarges what a rule can express. All TRAIN patch searches finish exactly: priorities change search order but final rewards coincide. This lack of TRAIN quality gain precedes distribution shift, so mismatch alone cannot explain it. Information repair has not demonstrated optimizer benefit.

Witness conditioning improves gate yield; relations-only programs retain stronger scalar fit and W costs are not lower. Warm deployment establishes no incremental reward gain over Degree or EoH. Only ground-gap changes are paired; small unit-weight TRAIN/TEST share resource regimes. Large tests jointly change scale, weight and density. They establish neither isolated configuration gain nor recurring large-scale decision semantics. Boundary-aligned evidence and matched large-scale training require further study. Renaming contradictions, Grids cohort failures, cold losses and detailed costs remain in the report.

Finite DAG representability ensures neither bounded-grammar fit nor transfer. Guarantees also apply to enumerated proposals; LLMs construct, not certify. Heterogeneous weights change observed distinctions [22]. Additive optimality and nominal targets differ from minimum runtime and equal computation. Corpus exposure, unmatched compute and identifier ties further delimit generalization claims.

7 Conclusion

Certified representation–rule refinement diagnoses missing information before rule repair. Sound completion bounds and quotient witnesses direct typed LLM proposals, complemented by a scoped additive-cost guarantee. Witness feedback improves observed gate yield; joint selection raises withheld fit within W. Frozen classical deployment separates these information benefits from optimizer quality, enabling targeted constraint adaptation.

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